

Gaussian elimination preserves row space

Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be rows of a matrix.

We compare the row spaces before and after each of the three types of e.r.o. as shown below.

$$\begin{aligned} 1. \quad & \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n\} \\ &= \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, c\mathbf{a}_i, \dots, \mathbf{a}_n\} \end{aligned}$$

$$\begin{aligned} 2. \quad & \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_i, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n\} \\ &= \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_j, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n\} \end{aligned}$$

$$\begin{aligned} 3. \quad & \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n\} \\ &= \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_i + c\mathbf{a}_j, \dots, \mathbf{a}_n\} \end{aligned}$$